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PLAN 372 Final

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Rail-Pedestrian Risk

Introduction & Motivation

From 2022-2024, there were over 76,000 civilian complaints of blocked train crossings reported to the Federal Railway Administration (FRA). This not only raises concerns over trains blocking traffic, but also poses risks to pedestrians. Nearly one quarter of those complaints mentioned pedestrians climbing under or over the stopped trains. This report explores how patterns in pedestrian climbing behavior relate to neighborhood characteristics, such as walkability, income, and delay severity. The goal is to identify where risk is most concentrated and how public data can inform infrastructure planning.

A major motivation of this analysis is to examine the potential of civilian-reported data as a tool for urban planning and infrastructure assessment. Although the Federal Railroad Administration (FRA) is not legally required to collect reports of blocked train crossings, it maintains a public-facing complaint portal as a means of tracking incidents across the country. This analysis investigates how individual, anecdotal complaints, which are typically dismissed as too fragmented for policy use, can be systematically aggregated and enriched with spatial, demographic, and environmental data to identify high-risk locations. By integrating government and open-source datasets, the study demonstrates how civilian input can reveal otherwise hidden trends and inform more equitable and targeted allocation of transportation resources.

Key Question

How do neighborhood walkability and demographic characteristics relate to pedestrian climbing incidents at blocked train crossings?

Literature Review

While there is limited scholarly research focused on the impact of blocked train crossings, investigative journalists from across the country have reported on heavily impacted local communities and also national trends. A 2023 report by *ProPublica* documented the dangerous consequences of extended rail blockages, particularly in low-income communities where pedestrians and schoolchildren are often forced to climb over stationary trains to reach school, work, or medical care. A particularly potent example from this article depicts 8-year-old Jeremiah and his mother facing the daily decision of whether it's worth crawling over or under a train to get to school. The publication of this story prompted public discourse and led to an increase in national and local discourse around blocked train crossings, proving the need for this type of analysis.

Data Description

This analysis draws on a combined dataset of blocked train crossing complaints collected by the Federal Railroad Administration (FRA) between 2022 and 2024, alongside environmental and demographic data from the U.S. Environmental Protection Agency (EPA) and the American Community Survey (ACS). The dataset of complaints included 76,742 individual reports submitted by civilians via the FRA's [blocked crossed portal](#). The core FRA dataset includes complaint-level information such as the location of the crossing, duration of the delay, and description of the immediate impacts observed. To support geographic and demographic analysis, this dataset was spatially joined with three additional sources: Crossing Inventory Data from the U.S. Department of Transportation, the EPA National Walkability Index, and the American Community Survey. Crossing Inventory Data from the U.S. Department of Transportation provides geographic coordinates and structural information for crossing locations. The EPA National Walkability Index assigned walkability scores at the census block level based on factors such as neighborhood design, distance to transit, diversity of land use, and employment mix and density. Demographic data from the American Community Survey (ACS), accessed using the

tidycensus R package, including metrics used in this analysis such as median household income, population, and households without access to a vehicle.

Data Processing

Following basic data cleaning and preprocessing, the three data sets were joined together. Geographic data from the U.S. DOT Crossing Inventory was joined on the Crossing ID field. This enabled the extraction of latitude, longitude, and structural attributes necessary for mapping and spatial linkage. The spatial merge introduced some complications due to non-unique IDs in the Crossing Inventory dataset, prompting careful filtering to avoid duplicate joins.

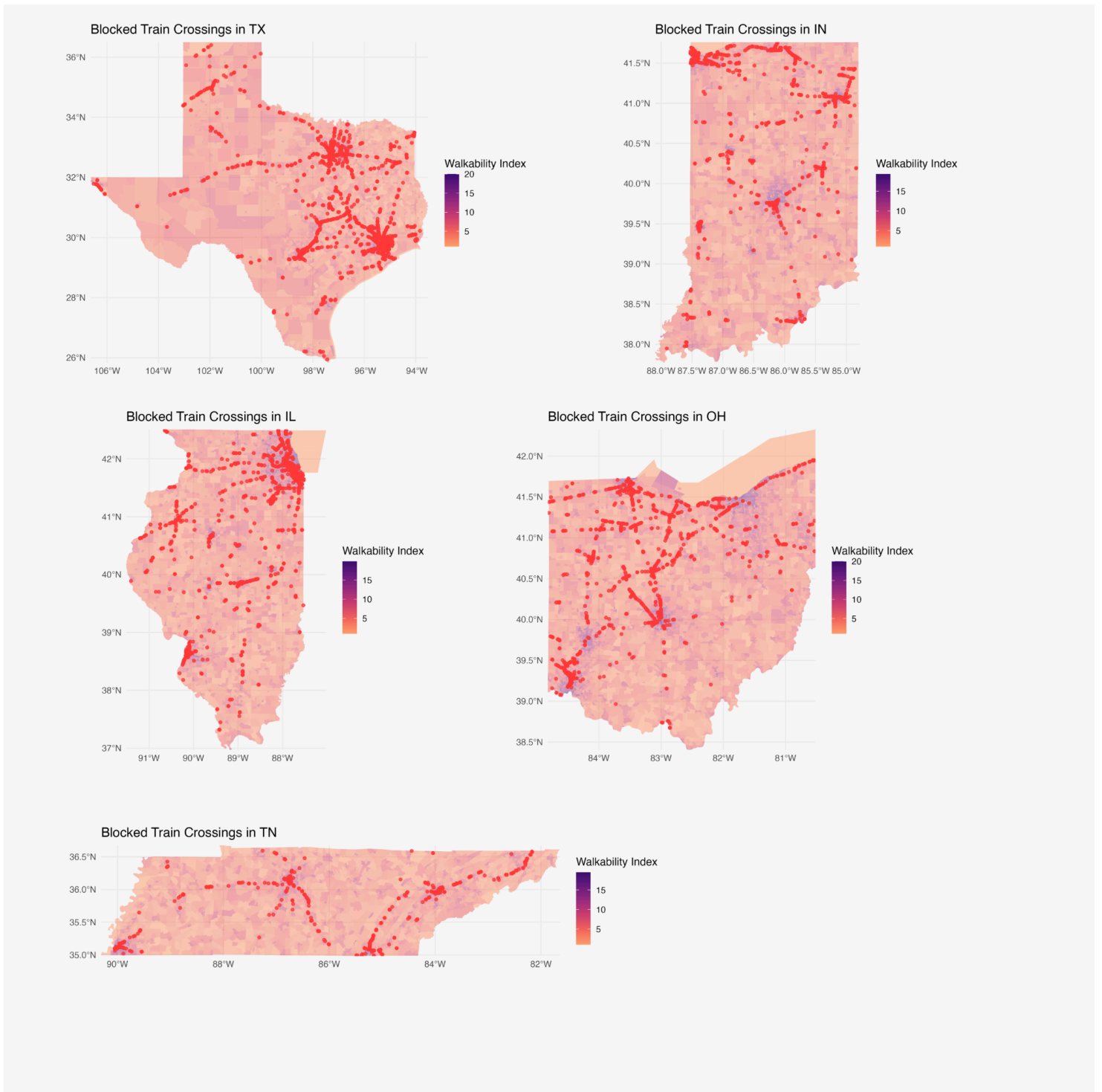
The EPA National Walkability Index and ACS demographic variables were integrated using spatial joins. Walkability scores were attached at the census block group level using the *sf* package. Demographic characteristics such as median income, population density, and percentage of households without a vehicle were acquired at either the block group or tract level using *tidycensus*. These variables were selected based on their theoretical relevance to pedestrian vulnerability and transportation equity.

Exploratory Analysis

Exploratory analysis revealed several key patterns. 89.3% of complaints involved delays longer than 15 minutes. Specific crossings in Houston, TX, such as Crossing ID 859522Y, were reported over 1,300 times, highlighting potential hotspots for infrastructure intervention. 23.4% of all complaints cited pedestrian climbing behavior, a central focus of the regression modeling that followed.

These patterns guided the decision to limit the regression sample to the five states with the highest complaint volumes—Texas, Illinois, Ohio, Indiana, and Tennessee—thus balancing geographic breadth with manageable data size and computational feasibility. These five states accounted for about half of total complaints in the dataset. After merging all relevant datasets and removing observations with missing geographic or demographic information, the final regression dataset consisted of 249 complete crossing-level observations.

Here is a preliminary mapping I did to show the blocked train crossings and the walkability scores for the top five most blocked states.



Methods & Results

Multivariable Linear Regression

To begin my analysis on my core question of pedestrian and blocked rail crossing interactions, I chose to construct a multivariable linear regression. I modeled the climbing percentage using predictors like walkability, income, population density, percent of households without cars, and if the delay exceeded 15 minutes.

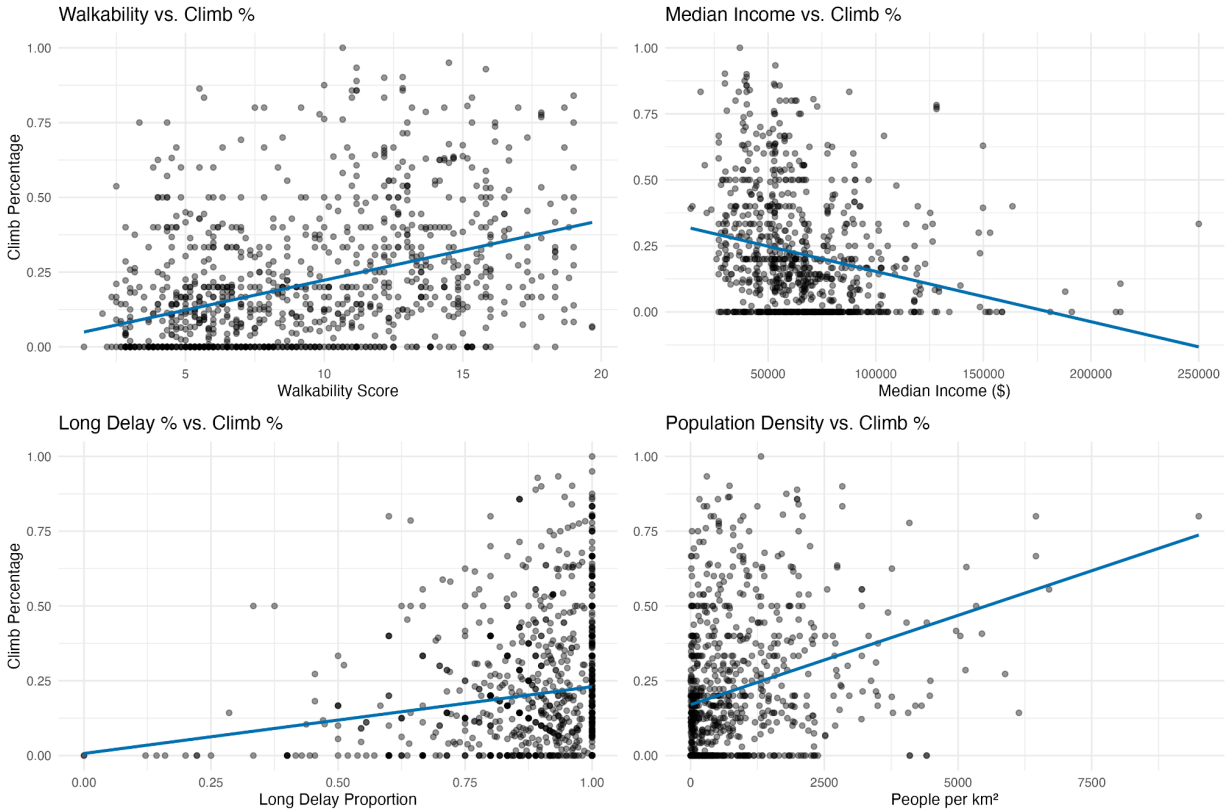
Outcome (dependent) variable:

- Percentage of complaints at a given crossing that involved pedestrians climbing on, over, or through the train cars

Independent variables:

- Walkability Index (EPA)
- Median Household Income (ACS)
- Population Density (ACS)
- Percentage of Households Without a Vehicle
- Proportion of Long Delays (percentage of complaints at a crossing that reported delays exceeding 15 minutes)

I first conducted bivariate exploratory analysis to see if there were any obvious trends between these proposed variables and the percentage of incidents at the train crossing that mention pedestrians climbing over or under these trains. The figure below presents scatterplots of Climb Percentage against each of the four main predictors used in the regression model: Walkability Score, Median Income, Long Delay Proportion, and Population Density.



All variables were normalized prior to regression to enable direct comparison of effect sizes. The modeling dataset was restricted to 249 observations representing crossings with full data coverage in the five states with the highest complaint volumes: Texas, Illinois, Ohio, Indiana, and Tennessee.

Multivariate Linear Regression Results

The model achieved an R^2 of approximately 0.22. While not suitable for forecasting, the model provides valuable insight into the environmental and social factors associated with pedestrian risk at blocked crossings. The results suggested that longer delays, higher walkability, and lower income levels were significantly associated with more frequent pedestrian climbing behavior.

Linear Regression Summary

term	estimate	std.error	statistic	p.value
(Intercept)	-0.136	0.055	-2.457	0.0141899
walkability	0.015	0.002	7.357	< 0.001
pop_density	0.000	0.000	3.102	0.0019837
median_income	0.000	0.000	-3.324	< 0.001
pct_no_vehicle	0.002	0.001	1.410	0.1589299
long_delay_pct	0.277	0.048	5.784	< 0.001

Determining High Risk Crossings

To complement the regression analysis and provide actionable insights for planners and policymakers, I developed a composite risk score to identify crossings where pedestrian climbing behavior is most prevalent and potentially dangerous. While the regression model revealed general associations between contextual factors and climbing risk, this scoring approach enables location-specific diagnosis which could be useful in real-world scenarios such as funding decisions for railway over and underpasses.

This scoring method is intended as a diagnostic tool, not a predictive one. The value is in its ability to find the most problematic crossings across the country based on factors that both observable pedestrian risk and underlying structural or socioeconomic vulnerabilities.

Construction of the Risk Score

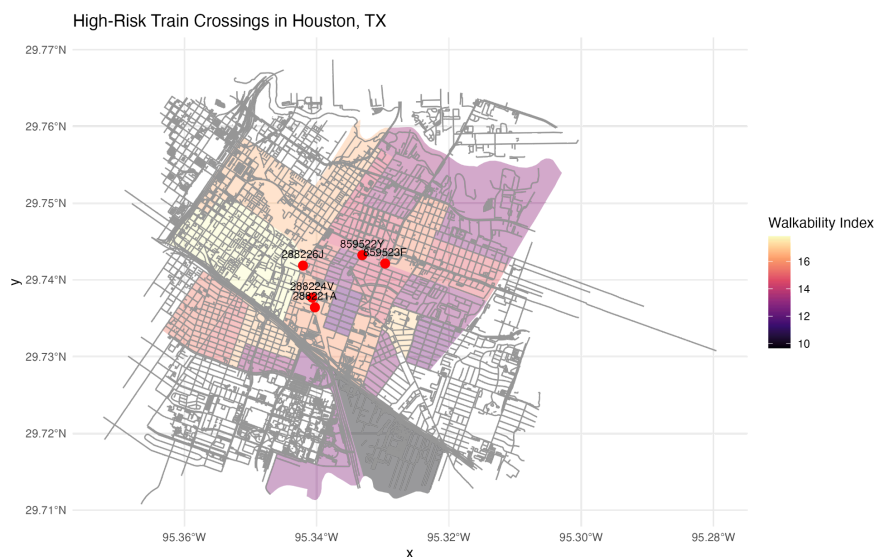
The composite risk score was calculated by combining five variables, each standardized (z-scored) to ensure comparability across different units and weighted in accordance with their perceived impact:

- Climbing Percentage (weight: 1.5)
- Walkability Index (weight: 1.0)
- Long Delay Percentage (weight: 1.0)
- Number of Total Complaints at Crossing (weight: 1.2)
- Median Household Income (weight: -0.5)

During the construction of these risk scores, crossings with fewer than five complaints were excluded to reduce the influence of outliers or rarely reported crossings.

Risk Score Results

Crossings were sorted in descending order of the computed risk_score, and the top 20 highest-risk crossings were selected for further inspection. All five of the crossings with the highest risk scores were in a neighborhood in East Houston in Texas.



This data confirmed what local journalists are already reporting. In December 2024, Houston local news outlet KPRC 2 reported on how residents of Houston's East End still were impacted heavily by blocked train crossings, even as officials said it was getting better. This shows promise in the risk score's ability to identify potential high risk crossings, then later verify the crossings picked by looking at local media and talking to local leaders.

Policy Implications

As reported by ProPublica, railroads are regulated at the federal level, which means that meaningful policy change and funding for infrastructure solutions, such as pedestrian overpasses, must be driven by federal agencies and congressional action. While local governments may be aware of high-risk crossings, they often lack the jurisdiction or resources to address them without federal support.

Analysis like these using citizen reported data shows potential for the further use of this data to identify problematic crossings and identify resources. By adapting the risk score to certain factors like reported pedestrian climbing rates, policy makers on the national level can choose which type of crossings they want to focus on and direct funding to the areas with the highest rate of these problems. Additionally, similar analysis techniques could be used by federal, state, and local governments to investigate other public data sources and civilian complaints. Top use cases could be for public trail and road conditions.

Limitations

This analysis is limited most significantly because of its reliance on citizen reported data. There's a sample selection bias on the people who submit these requests and what type of scenario would cause someone to report. There's also an awareness issue; some areas with more reports may know that they are more able to report blocked train crossings. Another limitation stems from the decision to restrict the regression analysis to five states based on their high complaint volumes. This choice allowed for more efficient data processing and ensured focus on locations with meaningful complaint density. However, it also limits the generalizability to other states or areas of the country.

In addition, datasets used to contextualize pedestrian risk like the EPA National Walkability Index and ACS demographic variables are reported at the block group or tract level, while crossings are point-based. This mismatch introduces a degree of spatial imprecision when assigning neighborhood characteristics to specific crossings, particularly in urban fringe areas or zones with irregular boundary geometry.

Lastly, while the regression model surfaced meaningful relationships, it accounted for just 22% of the variance in pedestrian climbing behavior. The modest explanatory power reflects both the complexity of pedestrian decision-making and the limitations of the available data. The composite risk score developed in the second stage of the analysis is intended as a diagnostic framework, not a predictive model, and its results are sensitive to the chosen variable weights and standardization methods.

Conclusion

This project illustrates the value of integrating civilian-reported data with spatial and demographic datasets to assess pedestrian risk at blocked train crossings. The analysis identifies clear associations between pedestrian climbing behavior and structural factors such as delay duration, walkability, and income level, particularly in high-risk urban areas like East Houston.

Although the regression model has weak explanatory power, the model still shows some evidence for longer delays, higher walkability, and lower income levels being significantly associated with more frequent pedestrian climbing behavior. The risk score demonstrates diagnostic utility of aggregated complaint data and the potential for composite risk scoring to inform federal infrastructure priorities. Broadly, this work shows how participatory data submitted by civilians can support targeted, evidence-based policy interventions in transportation planning.

References

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Data Sources

Environmental Protection Agency. (n.d.). EPA.

<https://www.epa.gov/smartgrowth/smart-location-mapping#walkability>

Safety, F. O. of R. (2025, April 24). *Crossing inventory data (form 71) - current: Department of Transportation - Data Portal*. DOT Open Data Catalog.

<https://data.transportation.gov/Railroads/Crossing-Inventory-Data-Form-71-Current/m2f8-22s6>

Blocked crossings. <https://www.fra.dot.gov/blockedcrossings/incidents>

GitHub Repository with source code

<https://github.com/aefatter/plan372-blocked-train-crossings>